Okonkwo Moses Chukwuka

Website | **Email**

PROFESSIONAL INTERESTS

I have a dedicated passion for the field of robotics, particularly computer vision for robotics. Over the years, I have specially made efforts to find solutions for robotic space understanding and on top of that, robotic space reasoning for downstream tasks. This has exposed me to different algorithmic paradigm like 3D gaussian splatting and SLAM algorithms in general. Allowing me to build substantial skills in python & C++. Though I believe computer vision is the door to other robotic solutions, my interests also include navigation, manipulation, and overall system optimization for edge devices.

PERSONAL DATA

Date/Place of Birth:	8th July, 1997, Enugu City, Enugu State.
Marital Status:	Single.
Nationality:	Nigerian.
Current Address:	Zhejiang University, No 38 Zheda Road, Xihu District, Hangzhou, Zhejiang, China, 310027

EDUCATION AND ACADEMIC QUALIFICATIONS

2016 - 2017**:**	International Exchange College, Xiangtan University, Hunan, China. **Chinese Language Proficiency. HSK4**
2017 - 2021:	College of Automation and Electronic Information Engineering, Xiangtan University, Hunan, China. **Bachelor of Engineering Degree in Electronic Information Engineering.**
2021-2024	College of Electrical Engineering**,** Shenyang University of Technology, Liaoning, China. **Masters** in Electrical Engineering: **Computer Vision for Robotics**
2024-Present	School of Electrical Engineering**,** **Zhejiang University, Zhejiang Hangzhou, China**. **PhD** in Electrical Engineering- **Computer Vision for Autonomous Systems.**
2025.4-2025.8	AutoLab, School of Engineering, **WestLake University,** Zhejiang Hangzhou, China. Visiting student under **Prof. Kaicheng Yu**

Professional References.

- Prof. Yang Jiaqiang
  - Relationship: PhD. Supervisor. Zhejaing University
  - Email: y.j.qiang@zju.edu.cn

- Prof. Junyou Yang
  Relationship MSc. Thesis Advisor. Shenyang Uni. of Tech.
  Email junyouyang@sut.edu.cn

REPRESENTATIVE PUBLICATIONS.

2024: **Moses Chukwuka Okonkwo**, Junyou Yang, Yizhen Sun, Guang Yang, Fausto Pedro García Márquez. RDSP-SLAM: Robust Object-Aware SLAM based on Deep Shape Priors. IEEE ACCESS, 12th, January, 2024.

2026: **Moses Chukwuka Okonkwo**, Yizhen Sun, Jiaqiang Yang. WINGS-SLAM: Wild Natural-Language Gaussian Splatting SLAM. [**Upgrading**], 2026

HONOURS, CERTIFICATES AND AWARDS

2017: Certificate of Proficiency, HSK level IV, Confucius Institute Headquarters, Hanban.

2017: Chinese Speech Contest for Foreigners in Xiangtan - Second Position.

2018: Hunan Government Scholarship for International Students, Hunan, China.

2019: Hunan Government Scholarship for International Students, Hunan, China.

2019: Xiangtan University Scholarship for International Students, Xiangtan, Hunan, China.

2021: Chinese Government Scholarship CSC, Shenyang University of Technology.

2024: Chinese Government Scholarship CSC, Zhejiang University of Technology.

WORK EXPERIENCE AND TRAINING

2019: Xiangtan University Summer Engineering Boot Camp, Embedded Systems and Programming in C.

2023 **Internship** - Perception Research Lab at Northeast University(东北大学):

a) Training of **crop row detetion deep learning** models
b) Training of **deep learing models for farm scene panoptic segmentation**
c) Deployment of deep learning models using **Pytorch** and **ROS** for **rubost visual navigation** of Agricultural Robots
d) Visual Navigation: ORB-SLAM(2/3).

MEMBERSHIP OF COMMITTEES

i. Member, Xiangtan University, Embedded System Development Lab - C programming. (2018 to 2021)

Engineering projects

(Please click the title to watch demo video)

2023: Robust Keyframe segmentation of visual Object-SLAM Based on Deepshape priors (ORBSLAM2, DeepSDF, YOLOv8/RTM-Det) [Ms Project]

2021: STM32 based mobile car with PID controlled motors[BSc Project].

2021: Servo motor PID speed control.

2020: Display of IMU and GPS sensor data using STM32 embedded system. [link 1] [link 2]

2020: Sound Detection Reading Glass for Deaf People, Blinks and vibrates when e.g the phone rings. [Team]

2020: Time based stepper motor control for dressing robot.

2019: 51 McU Line tracking mobile car based on photo-resistors

2019: Oscillator based Clock and Alarm using 8051 McU.

ENGINEERING FIELD (SKILLS).

i. C/C++ programming Language
ii. ROS
iii. Python(OpenCV, OpenMMLab, Pytorch)
iv. Linux(Ubuntu)[high]
v. Scientific Writing.

GENERAL INTERESTS.

i. Psychology
ii. Philosophy
iii. Project Management
iv. Entrepreneurship
v. Nutrition

Languages Spoken:

a. English Language – Very good and fluent.
b. Igbo – Fluent.
c. Mandarin – Fluent.